

MODEL QUESTION PAPER

TED (10) 5084

Reg. No.....

Signature.....

FIFTH SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY

INJECTION MOULD DESIGN

Maximum Marks: 100

Time: 3Hrs

PART-A

I (Answer all questions in one or two sentences, each question carries 2 marks)

- a) Write down the different types of cutting operations and its purposes.
- b) Why lathe is an unavoidable machine in injection mould making workshop?
- c) What are the advantages of guide pillars and bushes of an Injection mould ?
- d) State the empirical relationship for gate and its depth
- e) Write any two advantages of stripper plate ejection (5 x 2=10)

PART-B

(Maximum Marks: 30)

II (Answer any five of the following questions; each question carries 6 marks.)

- a) What are the various parts of a planning machine with the help of a neat sketch?
- b) Explain the working of a Cylindrical grinding machine.
- c) Distinguish between cavity and core of an injection mould with the help of a diagram and give its relationship for an integral mould.
- d) Explain the guiding and supporting ejector plate assembly for a part release system.
- e) What are the methods of cooling system for integer type injection mould for thermo plastics ?
- f) Compare the flat parting surface and stepped parting surface.
- g) How the products from fixed mould plates are ejected with the help of a diagram ? (5 x 6=30)

PART- C

(Maximum Marks: 60)

(Answer one full question from each module; each question carries 15 marks.)

Module-I

III How a rectangular cavity mould having 100x100x30xmm size cavity and two guide pins cylindrical cavity of size 10 mm bore size makes from a suitable metal plate ? (15)

OR

IV What are the machines and tools used for making mould in an advanced tool room with its specific purposes? (15)

Module-II

- V Draw a diagram of multi cavity mould for cup (8 Nos), mention its parts and purposes (15)

OR

- VI a) Explain the function of a register ring and guide pillars in an Injection mould (6)
- b) What are the different types of bolsters used in an injection mould, state the requirements for fixed type bolsters? (6)
- c) What are the advantages of pressure casting for mould making? (4)

Module-III

- VII Draw a neat sketch of the cross section of a two piece injection mould and label the parts, explain the ejector devises of that mould (15)

OR

- VIII a) What are the different methods of ejector techniques for removal of moldings from the mould? (10)
- b) Describe the inline ejector grid system and circular supported block grid system (6)

Module-IV

- IX a) What are the types of parting surfaces with examples ? (10)
- b) Explain the functions and runner , sprue , gate systems of an injection mould (6)

OR

- X a) Explain the positioning and balancing of a gate in a feed system (10)
- b) Identify and explain an ideal cooling system for a polypropylene bottle cap in a multi cavity injection moulding (6)
